

Technology #11: Gravity Compensation Hydraulic Damper for Postural Transients

Technology Domain: Hydraulic Valve Mechanics + Proportional Damping

Mechanism: CSF shunt with hydraulic damper that proportionally attenuates postural pressure transients. Unlike fixed anti-siphon devices (ASD) that switch open/closed, this provides continuous proportional damping based on gravity head.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Damper coefficient cannot be tuned to both attenuate transients AND maintain minimum drainage floor

Next Experiment: Damper element prototypes (various geometries)

Buyer Receives: Proportional damper concept + mechanism description; Gravity head + damper + orifice governing equations; Bench test protocols (c_h measurement, postural simulation, aging, C_{pk}); External precedent catalog (ASD literature: PMC9133390, Miethke ShuntAssistant); Critical UNKNOWN disclosure (optimal c_h , comparative advantage over ASD); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production damper element design (geometry, material, tolerances); Manufacturing process for damper + gravity reference + valve integration; Comparative clinical evidence vs existing ASDs; Regulatory submission (510(k) with substantial equivalence to ASD, or De Novo); Clinical validation (ICP excursion endpoint); Long-term reliability testing

Transfer Posture: Sponsored Validation

Target Buyer: Neuroshunt manufacturer; valve engineering company

CONFIDENTIAL. See full engineering dossier for details.